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valves should be slowly closed together, and a sample taken at the water discharge as the flow ceases (this point can be checked from the observation window).
1.2.11 A test identical to that described in 1.2.10, including the opening of the air cock, should be carried out with a mixture composed of approximately 25%. oil and 75%* water.
1.2.12 The equipment should be fed with 100%* of oil for at least 5 minutes during which time the observation window should be checked for any oil discharge. Sufficient oil should be fed into the equ~pment to operate the automatic oil discharge valve. After the operation of the oil discharge valve, the test should be continued for 5 minutes using a 100%* oil supply in order to check the sufficiency of the oil discharge system.
1.2.13 The equipment should be fed with water for 15 minutes. Samples of the separated water effluent are taken at the beginning of the test and after the first 10 minutes.
1.2.14 A test lasting a minimum of 3 hours should be carried out to check that the equipment will operate continuously and automatically. This trial should use a cycle varying progressively from water to oily mixture with

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first 10 minutes.
1.2.14 A test lasting a minimum of 3 hours should be carried out to check that the equipment will operate continuously and automatically. This trial should use a cycle varying progressively from water to oily mixture with approximately 25%* oil content and back to water every 15 minutes, and should test adequately any automatic device which is fitted. The whole test sequence
* percentage of volume
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should be performed as a continuous programme. At the end of the test, while the equipment is being fed with 25~~ oil, a water effluent sample should be taken for analysis.
1.2.15 Sampling should be carried out as shown in figure 2 so that the sample taken will suitably represent the fluid issuing from the water outlet of the equipment.
1.2.16 One litre narrow necked glass flasks utilizing caps with Teflon seals or equivalent should be used to collect the samples. Samples should be

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taken will suitably represent the fluid issuing from the water outlet of the equipment.

1.2.16 One litre narrow necked glass flasks utilizing caps with Teflon seals or equivalent should be used to collect the samples. Samples should be preserved with the addition of 5 ml of hydrochloric acid (see paragraph 4.3.1 of part 4 of the annex), unless the sample is to be extracted on the same day of collection, and be sealed and labelled in the presence of a representative of the national authority and arrangements should be made for anlaysis as soon as possible and in any case within seven days provided the samples are being kept between 2°C and 6°C at laboratories approved by the Administration.
1.2.17 The oil content of the samples should be determined in accordance with part 4 of the annex.
1.2.18 When accurate and reliable oil content meters are fitted at inlet and outlet of the equipment, one sample at inlet and outlet taken during each test will be considered sufficient if they verify, to within ± 10%, the meter readings noted at the same instant.
1.2.19 In the presentation of the results, the following data should be

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will be considered sufficient if they verify, to within ± 10%, the meter readings noted at the same instant.
1.2.19 In the presentation of the results, the following data should be reported, using the International Metric System of Units:
. 1 properties of the oil:
relative density at 15°C
viscosity (centistokes at 100°C/37.8°C)
flashpoint
ash
water content (total);
.2 properties of the water:
relative density at 15°C with details of any solid matter present;
.3 temperature at the inlet to the equipment;
.4 the method used in analysis of all samples taken and the results thereof together with meter readings where appropriate;
.5 a diagram of the test rig; and
.6 a diagram of the sampling arrangements.
1.2.20 The recommendations of the manufacturer of the filtering equipment concerning the choice and application of cleansing agents used for cleaning
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purposes in machinery spaces should be recorded in the appendix to the

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purposes in machinery spaces should be recorded in the appendix to the certificate of type approval. If the manufacturer declares that the user is free in the choice and application of cleansing agents because this does not affect the performance of the equipment, then this is to be recorded also. The manufacturer should include this information in the instruction manual for the filtering equipment.

PART 2 - TEST AND PERFORMANCE SPECIFICATIONS FOR TYPE APPROVAL OF OIL CONTENT METERS FOR BILGE ALARMS

2.1 General

2.1.1 These Test and Performance Specifications relate to oil content meters for bilge alarms. In addition, the electrical and electronic section of these systems should be tested in accordance with the Specifications for Environmental Testing contained in part 3 of this annex.

2.1.2 The meter being tested should comply with all the relevant requirements of the technical Specifications contained in section 4.2 of these Guidelines and Specifications.

2.2 Test specifications

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2.1.2 The meter being tested should comply with all the relevant requirements of the technical Specifications contained in section 4.2 of these Guidelines and Specifications.
2.2 Test specifications

2.2.1 For a meter designed for a bilge alarm , the accuracy should be within ± 5 ppm. The accuracy of bilge alarms should remain within the above limits despite the presence of contaminants other than oil, and the power supply varying by 10% from the design value, i.e. in respect of electricity, compressed air, etc.

2.2.2 The sampling arrangement should be such that a representative homogeneous sample is obtained under all conditions of operation and under all operational proportions of oil content. The sample should be obtained from the full flow through the meter, but when this is impracticable the sampling arrangements shown in figure 2 in part 1 should be used. Special care should be given to this stage of the process and the validity of the resultant findings.

2.2.3 During the various tests the response time of the meter should be checked and it should also be noted whether alarms operate adequately when a pre-stated threshold is exceeded.

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findings.
2.2.3 During the various tests the response time of the meter should be checked and it should also be noted whether alarms operate adequately when a pre-stated threshold is exceeded.

2.2.4 A diagrammatic arrangement of a test facility for evaluating the performance of oil content meters is given in figure 3. The accuracy of the oil content meter will be determined by comparing its readings against a known flow of oil injected into a known flow of water. The grab samples taken will

be analysed in a laboratory by the methods specified in part 4 of this annex. The results of the laboratory analysis will be used for correction and to indicate sampling and test equipment variability. The water flow rate will be adjusted so that the entire oil-water flow passes through the oil content meter, except the intermittent grab sample stream. Special care should be given to keep, continuously, a constant oil content in the water that flows into the meter. The oil and contaminant metering pumps should be adjusted to deliver a nearly continuous quantity of oil. If oil injection becomes
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into the meter. The oil and contaminant metering pumps should be adjusted to deliver a nearly continuous quantity of oil. If oil injection becomes
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